POWER BI D20

Time intelligence and SUMX

From the first comparison to a Nakheel overview page

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DATA SCIENCE & ANALYTICS

BY THE END OF TODAY, YOU CAN

Class objectives

01 Compare April revenue with other periods.

02 Diagnose incomplete periods and empty months.

03 Explain the rows and filters used by SUMX.

04 Choose and format charts for an overview page.

EVERYONE DO

Step 1.1

One report row supplies a filter

Which sales lines enter Completed Revenue on the Outerwear row?

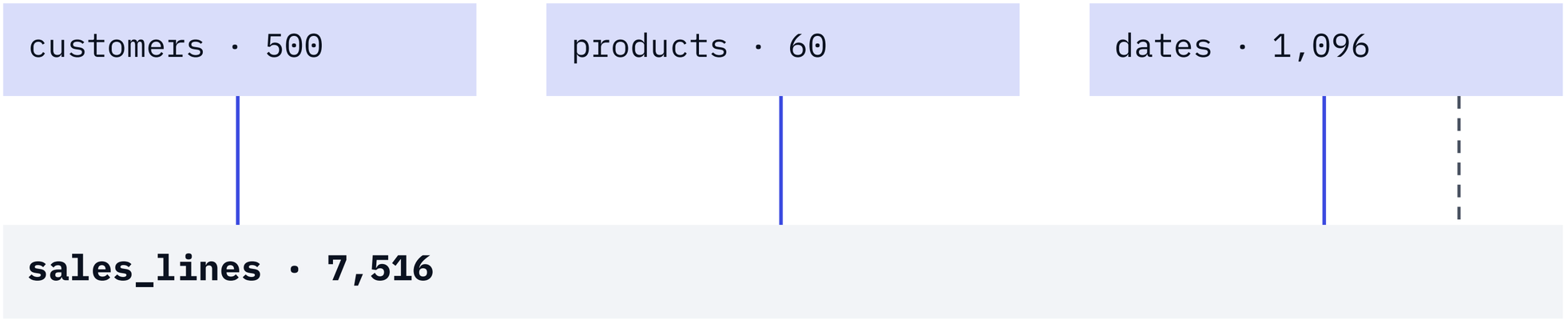
PART 1

The starting model

Four files and four reusable measure definitions

Model | Dates | Sales lines | Report

Four files give us the Nakheel star



The order-date path is active. After loading the four measures, Completed Revenue is 316,023.50.

EVERYONE DO

Step 1.2

Build the fresh model

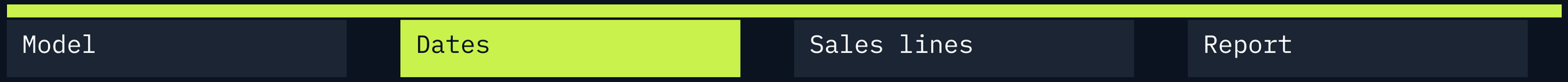
Load the four tables, mark the date table and paste the four measures.

INSTRUCTOR DO

PART 2

Revenue over time

One measure evaluated with different dates



NAKHEEL RETAIL

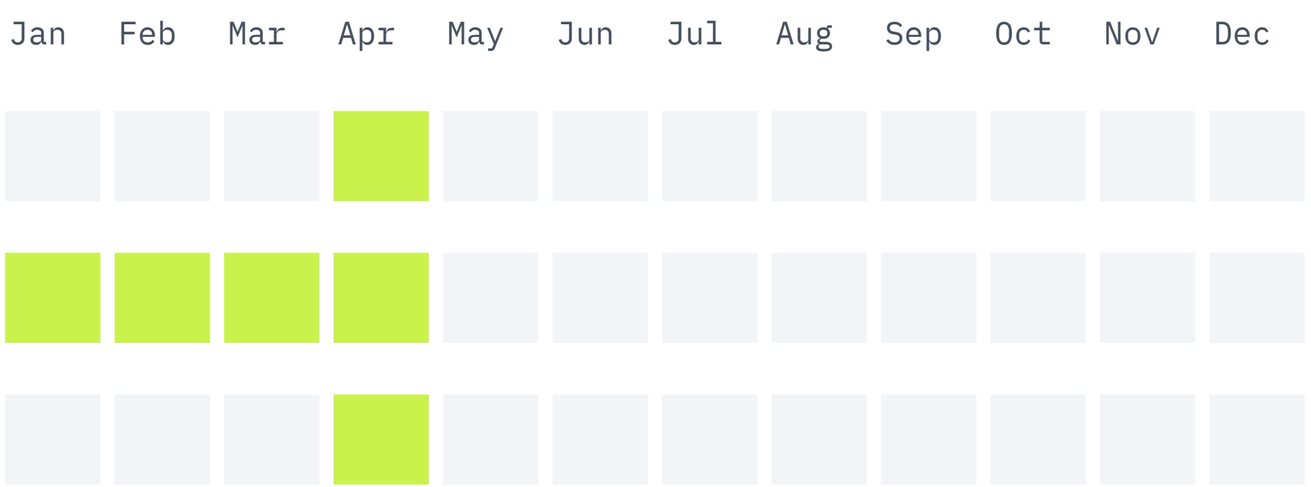
April revenue needs a comparison

April 2026	2026 through April	April 2025
20,237.00	**62,497.50**	**12,791.00**

The month, the year so far and the same month last year answer different questions.

NAKHEEL RETAIL

The April report row stays in place



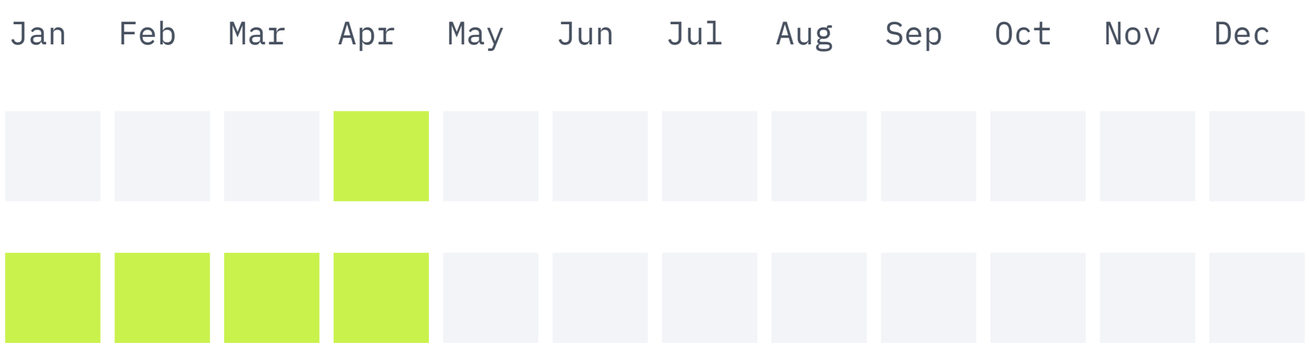
Each highlighted window supplies dates to the same Completed Revenue measure.

The date table supplies one row per day

Date	Calendar month	Active path
2026-04-01	April 2026	sales_lines[order_date]
2026-04-02	April 2026	sales_lines[order_date]
2026-04-03	April 2026	sales_lines[order_date]

Use unique dates without gaps, spanning full years. Mark the table. The dashed delivery-date path answers a different question.

Year to date starts on 1 January



April 2026	**20,237.00**
Jan–Apr 2026	**62,497.50**

For the April row, year to date includes 1 January through 30 April 2026.

DAX

TOTALYTD evaluates Completed Revenue

```
Revenue YTD =
TOTALYTD(
    [Completed Revenue],
    dates[date]
)
```

April 2026 · 62,497.50

The first argument supplies the measure to evaluate.

DAX

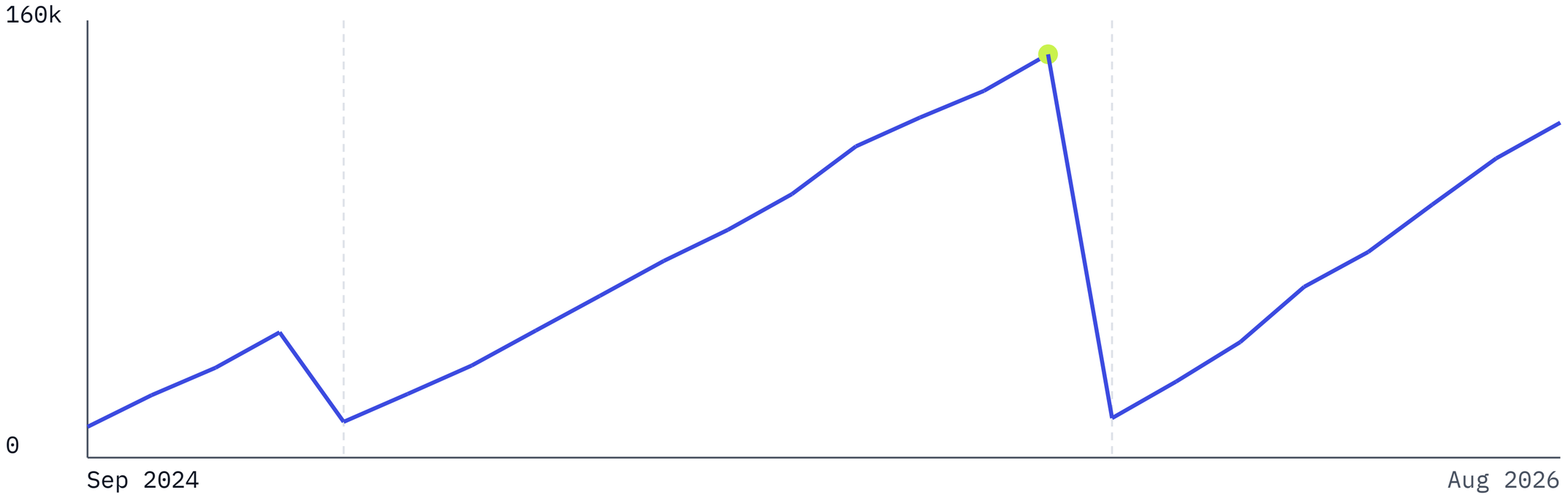
TOTALYTD uses the marked date column

```
Revenue YTD =
TOTALYTD(
    [Completed Revenue],
    dates[date]
)
```

The last date in the report cell’s filter sets the end of the window.

NAKHEEL RETAIL

Year-to-date revenue restarts in January



December 2025 reaches 147,589.00, the D10 result. January starts a new total.

EVERYONE DO

Step 2.2

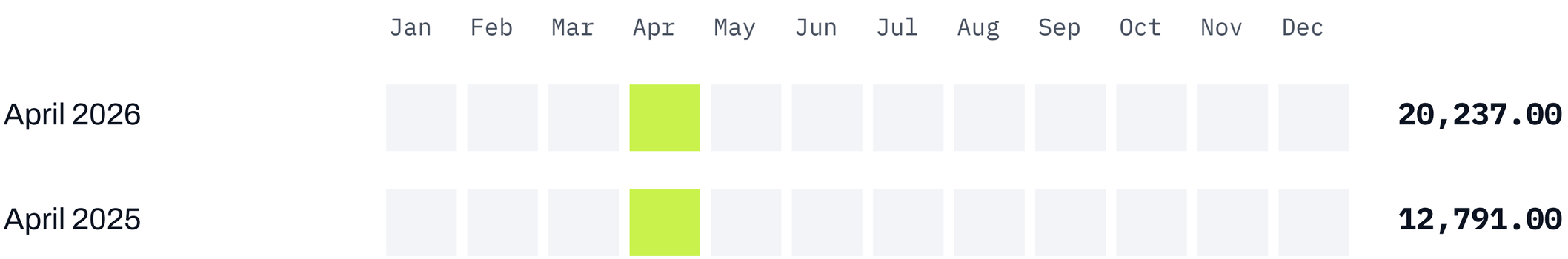
Write Revenue YTD

Check December 2025 against D10. The year-to-date result is 147,589.00.

INSTRUCTOR DO

NAKHEEL RETAIL

Last year uses the corresponding dates



The April 2026 cell can evaluate Completed Revenue over April 2025.

DAX

CALCULATE uses the shifted date set

```
Revenue LY =
CALCULATE(
    [Completed Revenue],
    SAMEPERIODLASTYEAR(dates[date])
)
```

April 2026 row · 12,791.00

SAMEPERIODLASTYEAR returns April 2025 dates. CALCULATE evaluates the measure with those dates.

DAX

YoY percent uses last year as its base

```
YoY % = DIVIDE(
    [Completed Revenue] - [Revenue LY],
    [Revenue LY]
)
```

Year-over-year (YoY) compares with the same period last year. Subtract last year’s revenue, then divide by it.

NAKHEEL RETAIL

April revenue is 58.2% above last April

April 2025	April 2026	YoY change
12,791.00	20,237.00	+58.2%

20,237.00 – 12,791.00 = 7,446.00. Divide that difference by 12,791.00.

DAX

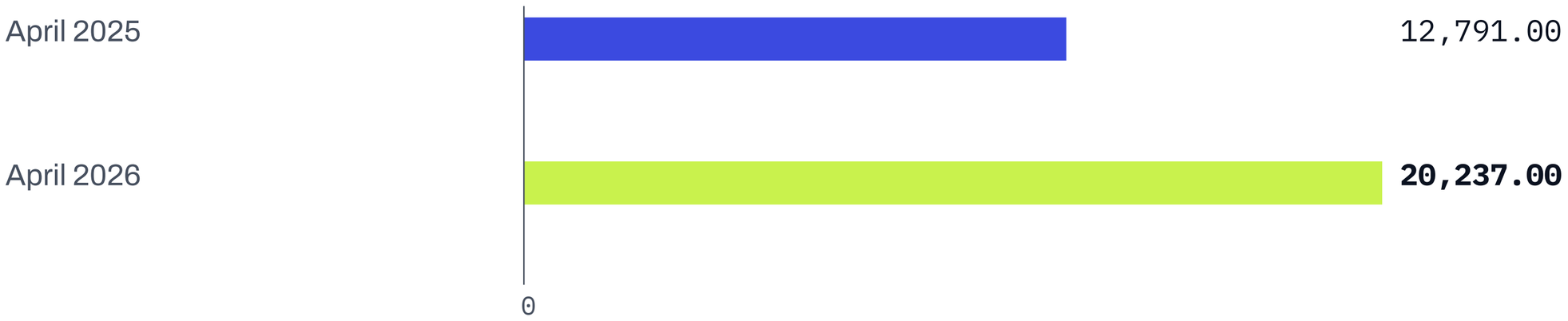
VAR gives the two values names

```
YoY % =
VAR CurrentRevenue = [Completed Revenue]
VAR PriorRevenue = [Revenue LY]
RETURN DIVIDE(
    CurrentRevenue - PriorRevenue, PriorRevenue
)
```

+58.2%

Each variable holds the result in this cell. RETURN uses those values to calculate the percentage.

The increase is measured against April 2025



The extra 7,446.00 is 58.2% of April 2025 revenue.

EVERYONE DO

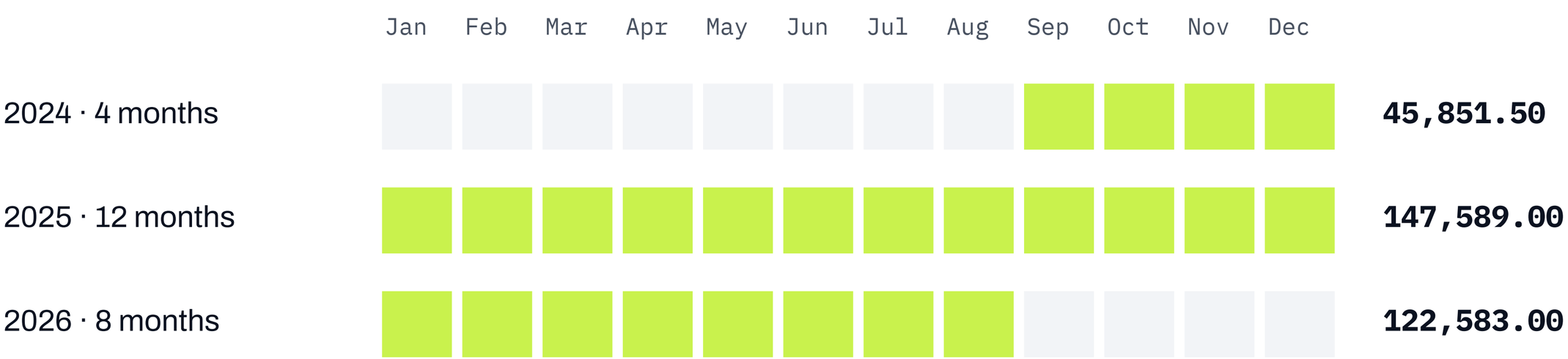
Step 2.4

Write Revenue LY and YoY percent

April 2026 returns 12,791.00 for last year and +58.2% for the change.

NAKHEEL RETAIL

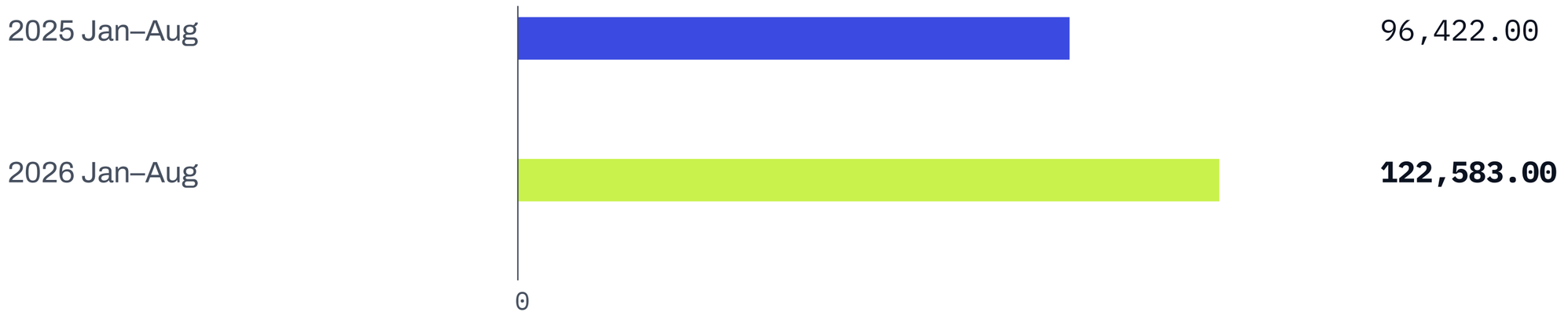
Calendar years can cover different months



2025 vs 2024 gives +221.9%. 2026 vs 2025 gives –16.9%. The month counts differ.

NAKHEEL RETAIL

The same eight months show a rise



January–August revenue rose 27.1% from 2025 to 2026.

NAKHEEL RETAIL

An empty month can display a 100% fall

Sep 2026	Sep 2025	Unguarded YoY
BLANK	17,472.00	-100%

DAX treats the current blank as zero in the subtraction. The snapshot ends on 31 August. Filter through that date.

PREDICT

Which comparison supports a growth claim?

Full-year labels	January–August	September row
−16.9%	**+27.1%**	**−100%**

Which value should appear in a summary of 2026 performance through August?

ANSWER

January–August is the comparable period

8 months vs 12	8 months vs 8	September 2026 missing
−16.9%	**+27.1%**	**−100%**

Only the middle value compares observed revenue over the same months.

EVERYONE DO

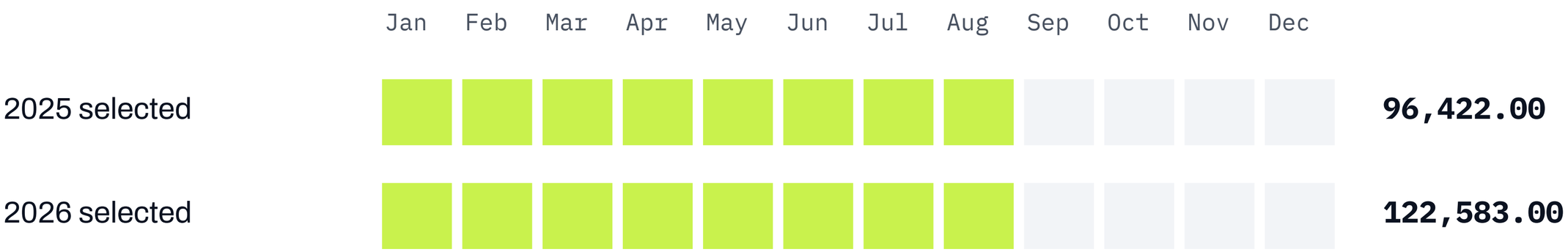
Step 2.6

Compare January through August

Filter both years to months 1 through 8. The 2026 change is +27.1%.

PERIOD CHECK

Check both date windows before reporting YoY



Match day coverage too. If the current snapshot ends on 15 August, compare through 15 August in both years.

INSTRUCTOR DO

PART 3

Revenue by sales line

Calculate each amount before adding the lines

Model | Dates | Sales lines | Report

DAX

The measure needs one quantity and one price

```
Revenue attempt =
    sales_lines[qty] * sales_lines[price]
```

This measure has no current sales line. Use an iterator to pair quantity with price.

NAKHEEL RETAIL

Multiplying column totals gives 288.00

Order 100142	Quantity	Unit price	Line amount
Hooded sweatshirt	1	18.00	18.00
Compression top	3	16.00	48.00
Yoga mat	2	14.00	28.00

6 × 48.00 = 288.00

The column totals mix prices and quantities from different sales lines.

A calculated column has a current row

Product	Category	Quantity	Unit price	Amount
Hooded sweatshirt	Outerwear	1	18.00	18.00
Compression top	Sportswear	**3**	**16.00**	**48.00**
Yoga mat	Sportswear	2	14.00	28.00

In row context, DAX reads values from the current line. The calculated column stores 3 × 16.00 = 48.00.

NAKHEEL RETAIL

SUMX evaluates an expression on each row

Product	Quantity	Unit price	Amount
Hooded sweatshirt	1	18.00	**18.00**
Compression top	3	16.00	**48.00**
Yoga mat	2	14.00	**28.00**

SUMX uses the quantity and unit price from each line to work out its amount.

The three line amounts add to 94.00

Hooded sweatshirt	Compression top	Yoga mat
18.00	**48.00**	**28.00**

18.00 + 48.00 + 28.00 = 94.00

An iterator runs an expression for each selected row. SUMX adds the results.

Filters select the lines SUMX visits

Order 100142	Category	Line amount	Outerwear filter
Hooded sweatshirt	Outerwear	**18.00**	**Included**
Compression top	Sportswear	48.00	Excluded
Yoga mat	Sportswear	28.00	Excluded

The Outerwear report row supplies filter context. Only its 18.00 line remains for SUMX to evaluate.

The first SUMX argument supplies the table

```
Revenue (SUMX) = SUMX(
    sales_lines,
    sales_lines[qty] * sales_lines[price]
)
```

In this measure, sales_lines contains the rows available under the current filters.

DAX

The second argument reads the current line

```
Revenue (SUMX) = SUMX(
    sales_lines,
    sales_lines[qty] * sales_lines[price]
)
```

Multiply quantity and price for the current line. SUMX then adds those amounts.

SUM and SUMX agree on these sales

Filter	SUM of line_value	SUMX of qty × price
All statuses	340,370.00	340,370.00
Completed	**316,023.50**	**316,023.50**

Use SUM for a supplied line amount. Use SUMX when the row expression needs to be calculated.

CALCULATE can turn a current row into a filter

Current order ID	Without CALCULATE	With CALCULATE
100003	141.00	**47.00**
100142	141.00	**94.00**

The report selects both orders. Context transition applies the current order ID as a filter so each sum uses that order’s lines.

A slicer changes the measure result

Calculation location	Result	What it describes
Source or Power Query	48.00	Compression top line
DAX calculated column	48.00	Compression top line
Model measure	**94.00 → 18.00**	Whole order → Outerwear only

A stored line value supports row-level use. A reusable measure supplies the total under the report filters.

EVERYONE DO

Step 3.2

Write Revenue SUMX

For order 100003, reproduce 141.00 from the wrong formula and 47.00 from SUMX.

A quick measure generates a formula

Calculation	Base value	Date field
Year-to-date total	**Completed Revenue**	**dates[date]**

Choose Year-to-date total and Completed Revenue. Open the generated DAX to see which date field it uses.

NAKHEEL RETAIL

A supplied amount can be summed directly

Supplied line_value

48.00

Quantity × price

48.00

SUM reads the supplied amount. SUMX works out the expression on each line.

NAKHEEL RETAIL

The two revenue measures can be reconciled

Order 100003	All statuses	Completed only
47.00	**340,370.00**	**316,023.50**

If the totals differ, match the filters and compare the amounts on individual sales lines.

Break

Next, the Nakheel overview

INSTRUCTOR DO

PART 4

The first report page

Choose the picture that answers the question

Model | Dates | Sales lines | Report

NAKHEEL RETAIL

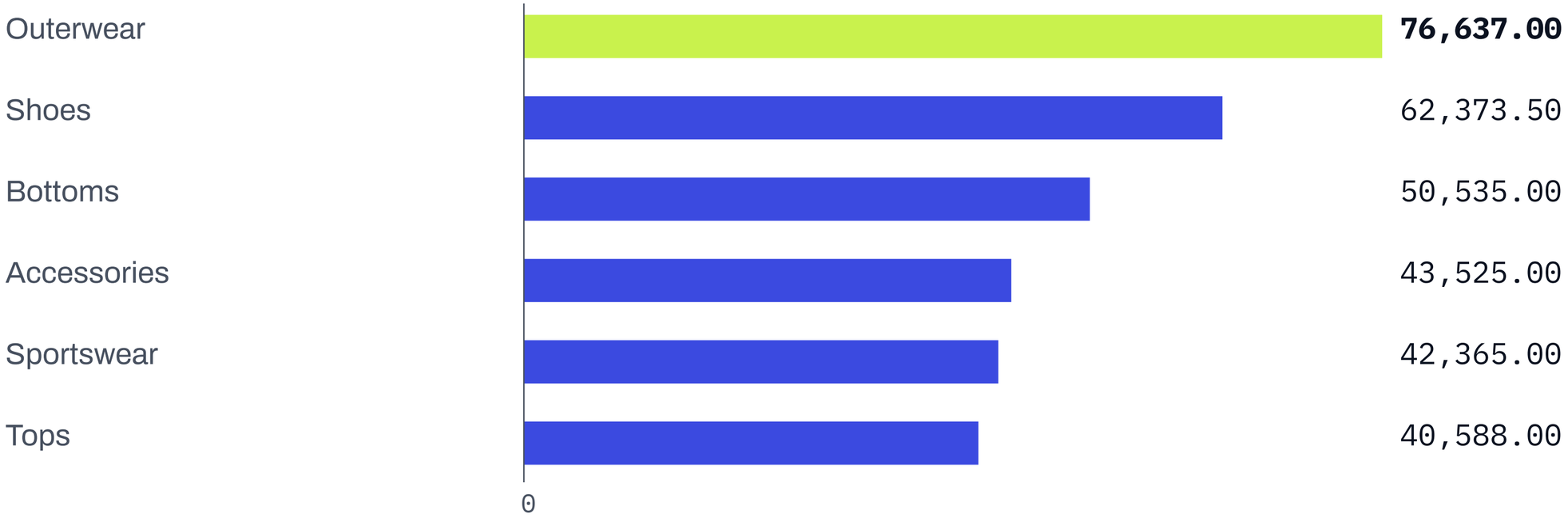
The question determines the visual

Question about Nakheel	Visual	What you read
Which category brings in most revenue?	**Bars**	Lengths on one scale
How does revenue move each month?	**Line**	Change through time
What order values are common?	**Histogram**	Orders in each value band
Do units sold vary with price?	**Scatter**	A pattern across products

The same data can answer several questions. Give each visual one job.

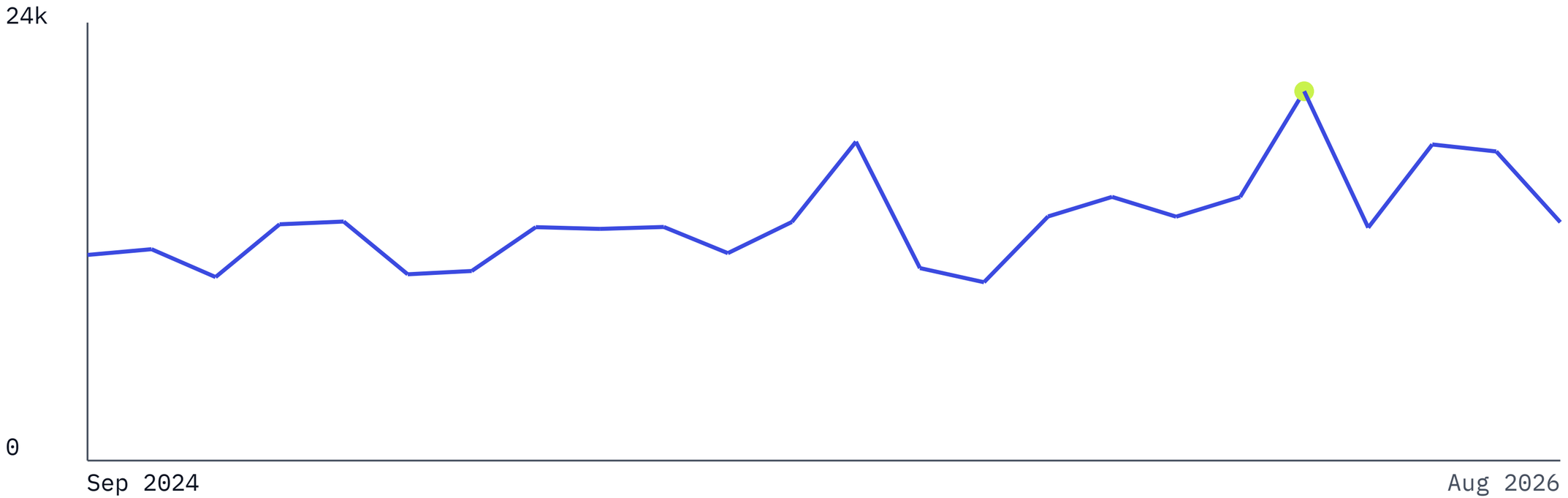
NAKHEEL RETAIL

Sorted bars compare category revenue



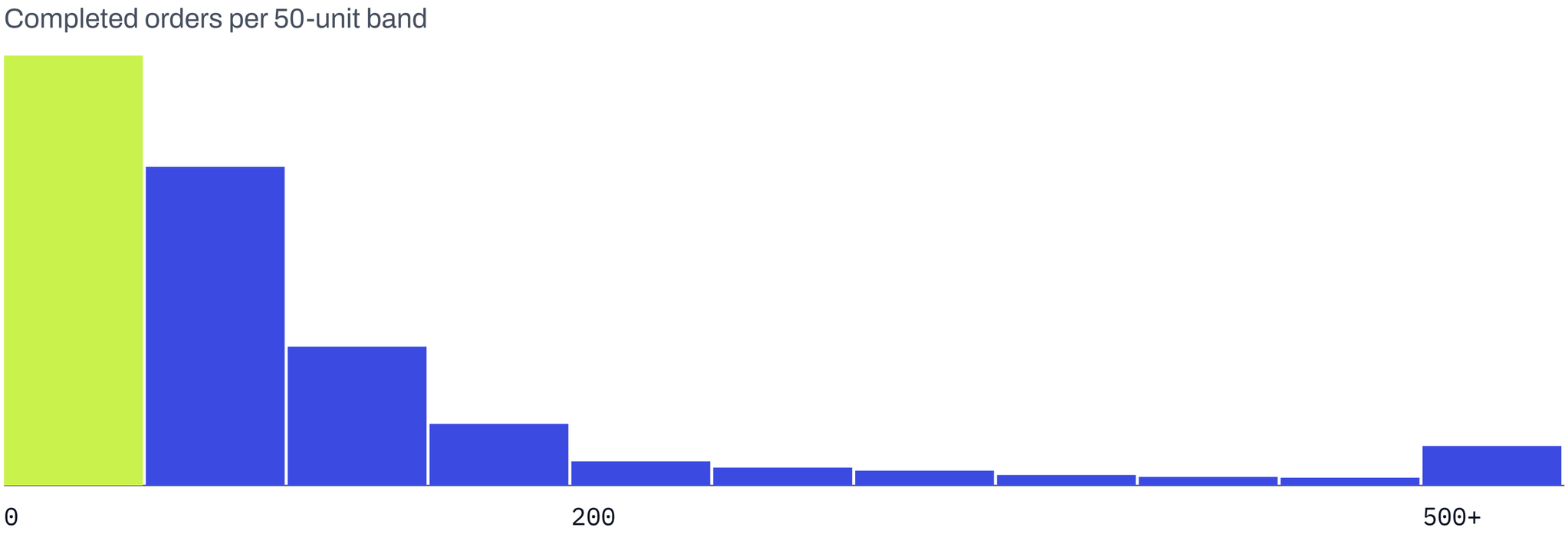
Outerwear brings in 76,637.00. All bars start at zero.

A line shows monthly revenue in order



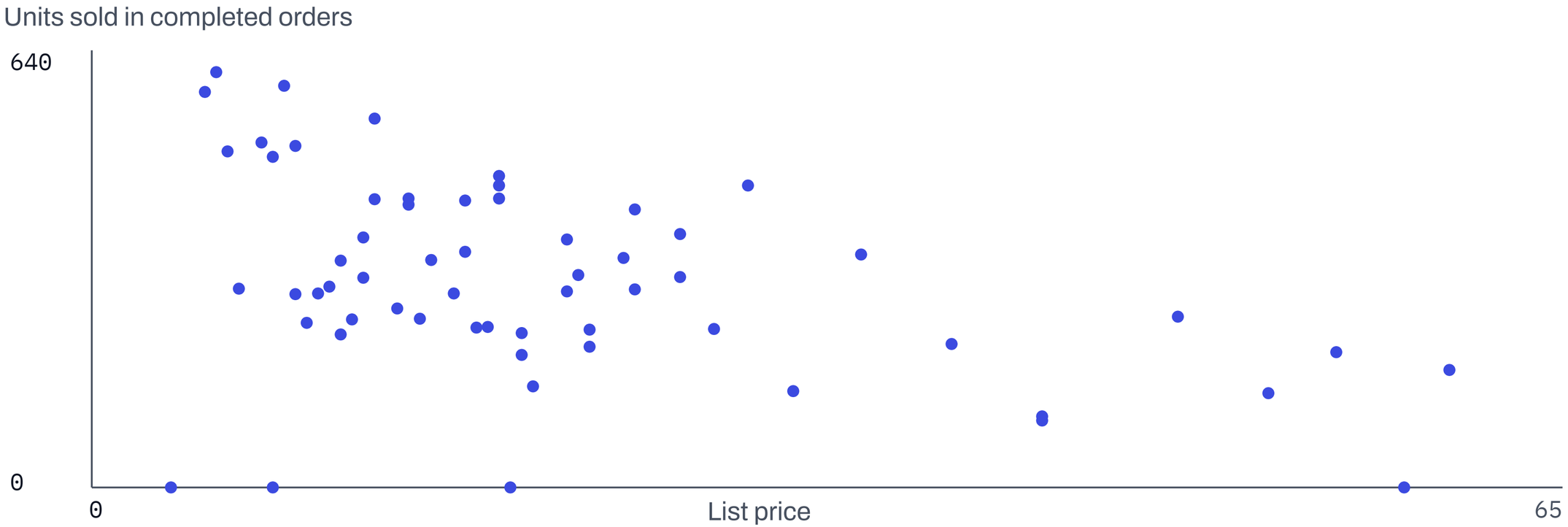
April 2026 is the highlighted point at 20,237.00. Each point sums one month.

A histogram counts orders by value band



The first band counts 1,112 orders below 50. The last bar pools orders worth 500 or more.

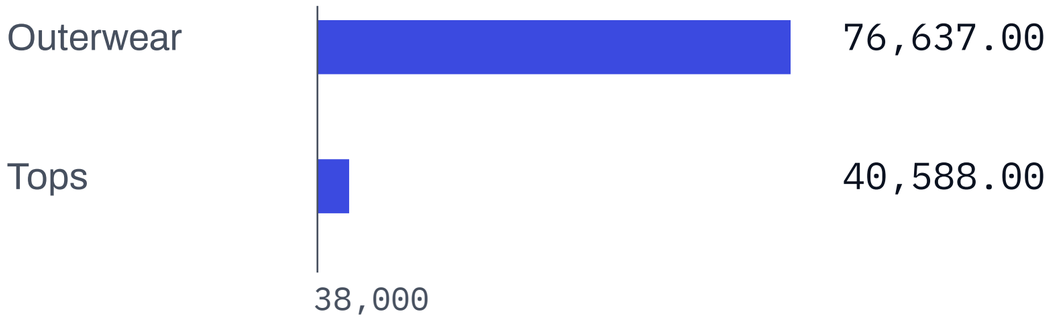
A scatter plot pairs price and units sold



Each dot is a product. The correlation is -0.45: higher prices tend to accompany fewer units sold. The pattern alone cannot explain demand.

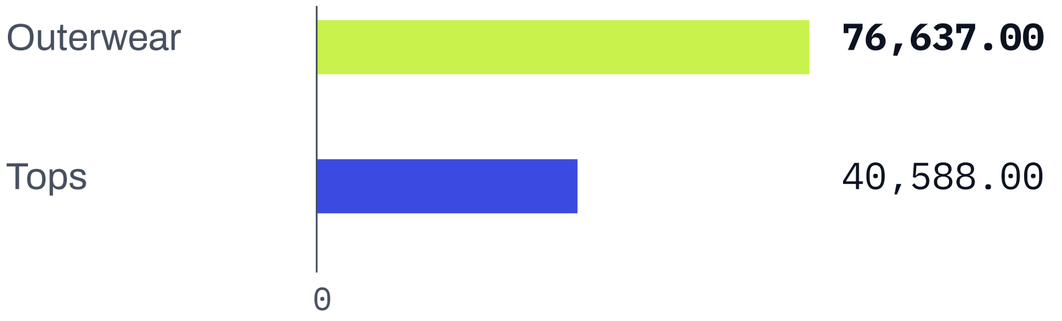
A truncated axis exaggerates the difference

Baseline 38,000



Length ratio 14.9×

Baseline zero

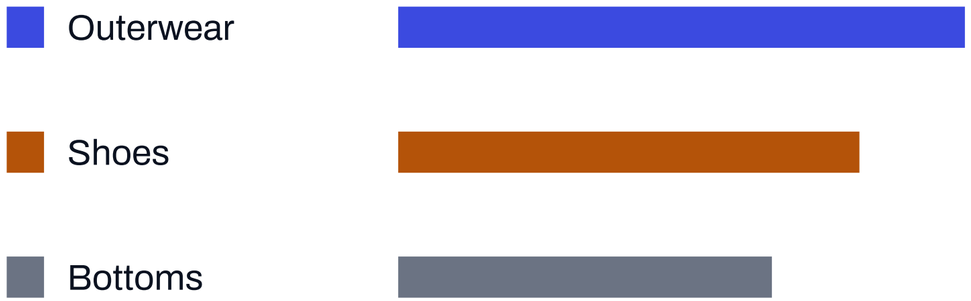


Length ratio 1.9×

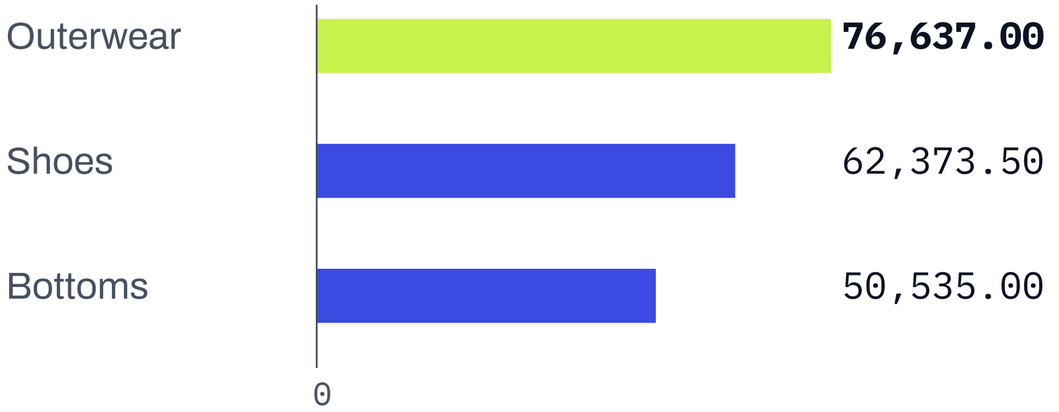
Start these bars at zero so their lengths represent the revenue amounts.

Direct labels make the comparison easier

Read a separate key



Read the category names beside the bars



Keep category names beside the values. Reserve color for the bar you want the reader to notice.

A theme sets report formatting defaults

Before theme



Nakheel theme



A theme changes default colors and styles. Measures, filters and visual overrides stay as you set them.

A formatting rule marks the sign of YoY

Order month	Completed revenue	Last year	YoY %
Apr 2026	20,237.00	12,791.00	**+58.2%**
Aug 2026	13,061.50	13,075.50	−0.1%

Color positive values one way and negative values another. Keep the sign visible and check the dates first.

EVERYONE DO

Step 4.2

Build the Nakheel overview

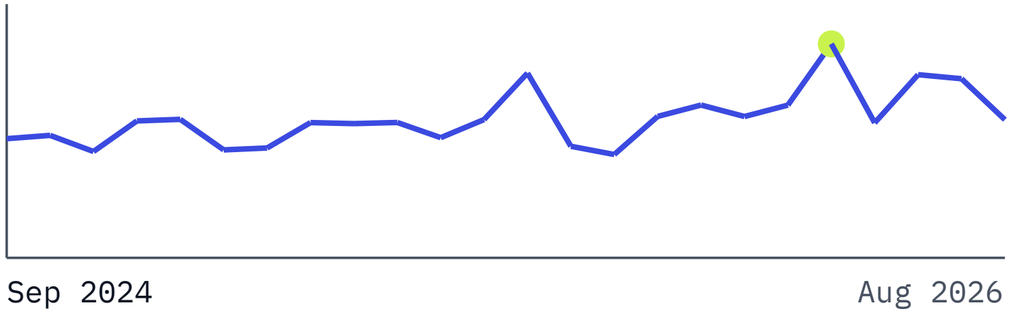
Apply the theme, add the cards and charts, and format the monthly comparison.

NAKHEEL RETAIL

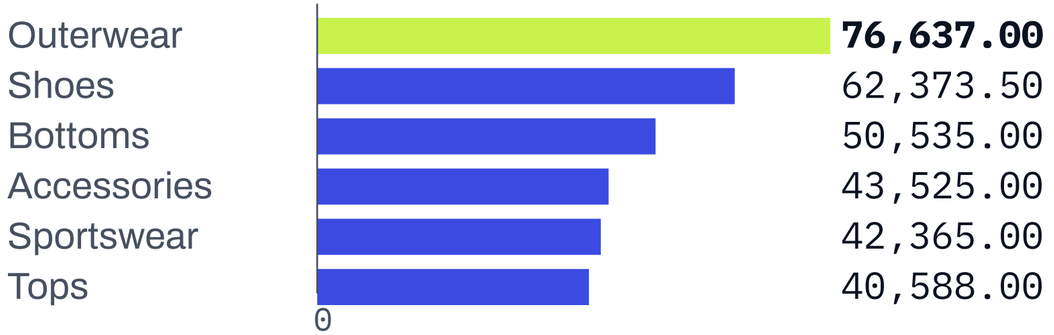
The overview compares months and categories

Completed revenue	Completed orders	Average order value
316,023.50	2,771	114.05

Revenue by order month



Revenue by category



All cards and charts use completed sales through August 2026.

INSTRUCTOR DO

NAKHEEL RETAIL

The cards need the same status filter

Orders · all statuses

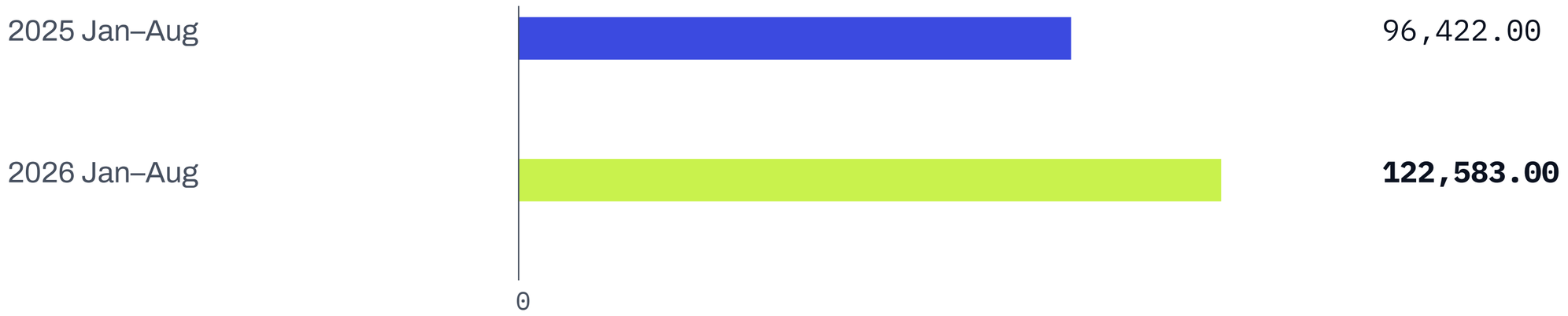
3,000

Orders · completed

2,771

Apply completed to the page so Orders and Average Order Value use the same population as Completed Revenue.

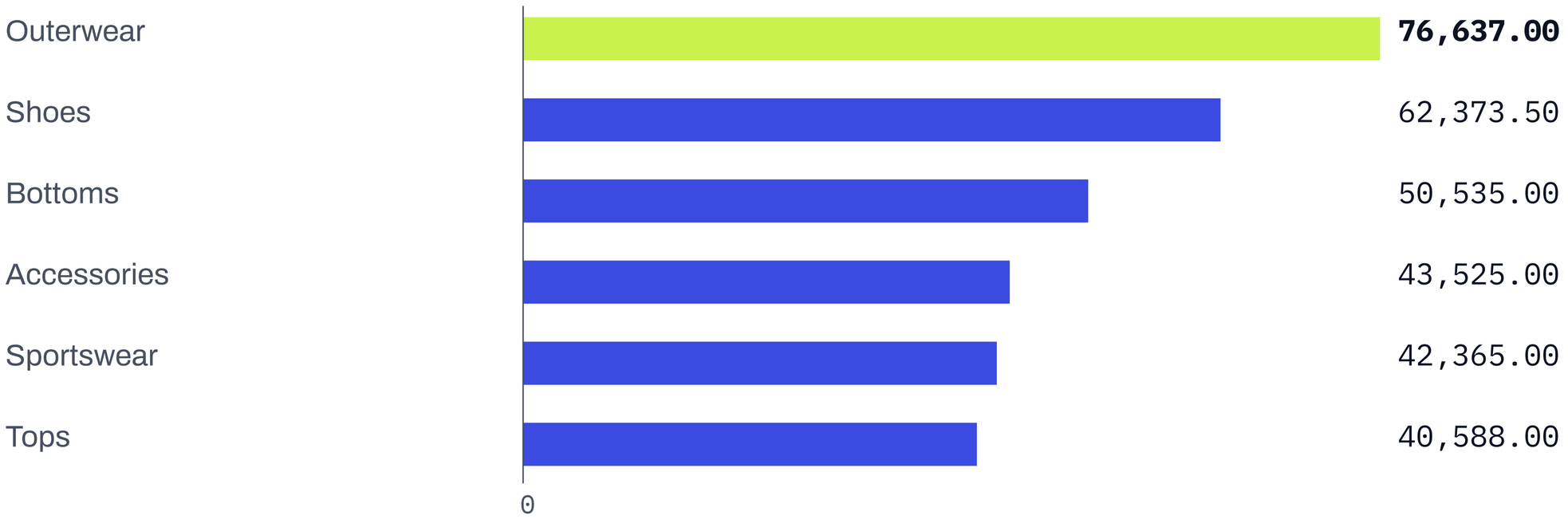
A report claim names its period



Completed revenue for January–August rose 27.1% from 2025 to 2026.

NAKHEEL RETAIL

The category chart covers the full snapshot



Outerwear leads across the snapshot. To explain the recent rise, compare the same months by category.

NAKHEEL RETAIL

Read the overview before you share it

What to check	What the reader should find
Date scope	Through 31 August 2026
Status scope	**Completed**
Monthly trend	**April 2026 · 20,237.00**
Category comparison	Outerwear · 76,637.00

Confirm these values and use pages 63–76 to study the calculations behind the report.

REQUIRED STUDY · DAX

Return BLANK for a month without revenue data

```
YoY % (with current data) =
IF(
    ISBLANK([Completed Revenue]),
    BLANK(),
    [YoY %]
)
```

Sep 2026 · BLANK

September 2026 has no current amount, so IF returns BLANK. A recorded zero still gives −100% against positive prior revenue.

Date functions return sets of dates

```
CALCULATE([Completed Revenue], DATESYTD(dates[date]))

CALCULATE([Completed Revenue],
    DATEADD(dates[date], -1, YEAR))
```

Jan–Apr 2026 · April 2025

On April 2026, these sets give 62,497.50 and 12,791.00. Classic DATEADD needs a contiguous selection.

CALCULATE applies the current order as a filter

```
SUMX(VALUES(sales_lines[order_id]),
    SUM(sales_lines[line_value]))        // 282.00

SUMX(VALUES(sales_lines[order_id]),
    CALCULATE(SUM(sales_lines[line_value]))) // 141.00

SUMX(VALUES(sales_lines[order_id]), [Total Revenue])

                                  Order 100142 alone · all three give 94.00
```

For orders 100003 and 100142, bare SUM repeats 141.00. CALCULATE gives 47.00 + 94.00. The measure reference does context transition automatically.

REQUIRED STUDY · DAX

FILTER keeps months above a revenue threshold

```
CALCULATE(
    [Completed Revenue],
    FILTER(VALUES(dates[year_month]),
        [Completed Revenue] > 15000)
)
```

April 2026 · 20,237.00

FILTER tests the measure each month. A Boolean CALCULATE argument cannot use a measure. From January–April, only April exceeds 15,000.

REQUIRED STUDY

A slicer leaves stored line values unchanged

Location	Expression or field	When	Line value
Source CSV	line_value	Load as supplied	**48.00**
Power Query	[qty] * [price]	Load or refresh	**48.00**
DAX column	sales_lines[qty] * sales_lines[price]	Model processing	**48.00**

When Power BI imports the data, it stores these line values. A slicer changes which lines enter the report total.

REQUIRED STUDY

RELATED retrieves the current line’s category

Current sales line	Matching product key	Category returned
Compression top	**P-057**	**Sportswear**

```
Category on line = RELATED(products[category])
```

RELATED follows the existing many-to-one relationship from this sales line to its product row.

REQUIRED STUDY

A DAX expression can create a model table

Input calendar	Month Index rows
1,096 day rows	36 distinct months
1 January 2024	2024-01
1–30 April 2026	**2026-04**
1–31 December 2026	**2026-12**

```
Month Index = DISTINCT(dates[year_month])
```

Power BI builds and stores this table in the model. Selecting a month in this disconnected table leaves sales unchanged.

A visual calculation uses the visual’s fields

```
YoY in this visual = DIVIDE(
    [Completed Revenue] - [Revenue LY],
    [Revenue LY]
)
```

April 2026 · +58.2%

Both fields must be in this visual. Removing Revenue LY removes an input. A model measure is reusable across visuals.

Closing stock uses one snapshot date

```
Closing Stock =
VAR ClosingDate = MAX(Inventory[SnapshotDate])
RETURN CALCULATE(
    SUM(Inventory[UnitsOnHand]),
    Inventory[SnapshotDate] = ClosingDate
)
// The date table filters Inventory[SnapshotDate]
```

Jan 12 Feb 9 Mar 14

Semi-additive: sum items at one date. For these made-up snapshots use March’s 14 units. Adding months gives 35. Empty April returns BLANK.

A calculation group applies a date window

```
Current = SELECTEDMEASURE()
YTD = CALCULATE(SELECTEDMEASURE(),
    DATESYTD(dates[date]))
Last year = CALCULATE(SELECTEDMEASURE(),
    SAMEPERIODLASTYEAR(dates[date]))
```

Current YTD Last year

A calculation item is a named formula applied to the selected measure. Put the item names on matrix columns.

REQUIRED STUDY

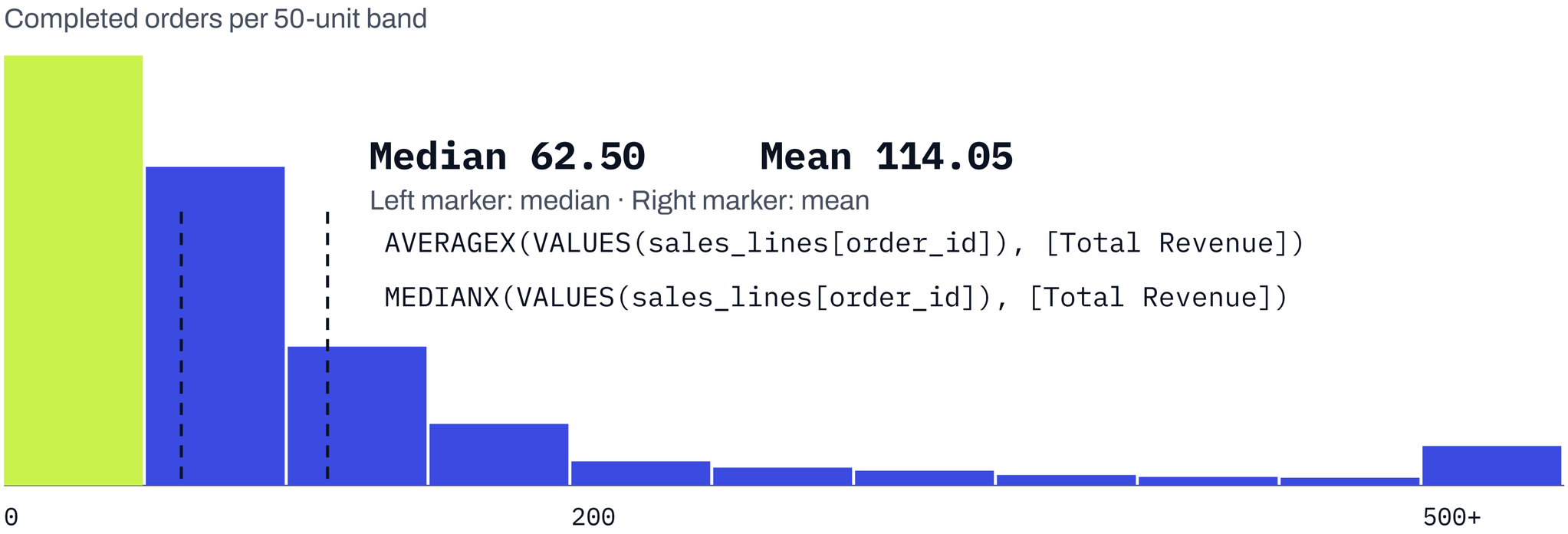
Revenue and orders use the same calculation group

April 2026	Current	Year to date	Last year
Completed Revenue	20,237.00	62,497.50	12,791.00
Orders · completed	**137**	**539**	**103**

Create the group in Model view. Enable Discourage implicit measures. Put the month on Rows, items on Columns and an explicit measure in Values.

REQUIRED STUDY

The median order value is below the mean



Filter status to completed. Each observation is one order. A few larger orders raise the mean.

REQUIRED STUDY

Counts of lines and orders answer different questions

Order 100003	Function	Result
Sales lines	COUNTROWS(sales_lines)	3
Order IDs	DISTINCTCOUNT(sales_lines[order_id])	**1**
Lowest line amount	MIN(sales_lines[line_value])	5.00
Highest line amount	MAX(sales_lines[line_value])	24.00

COUNT ignores blanks in a column. To count orders here, count distinct order IDs.

REQUIRED STUDY

Check a generated YTD measure against your own

Dialog selection	Value
Calculation	Year-to-date total
Base value	Completed Revenue
Date	**dates[date]**

Expected YTD · December 2025 147,589.00 · April 2026 62,497.50

Compare both results with the handwritten measure. Use that measure if the quick-measure control is unavailable.

STUDY INDEX

Required study and practice

63–66 · Dates and filters

67–70 · Calculation locations

71–73 · Stock and calculation groups

74–76 · Statistics and quick measures

Microsoft PL-300 study guide

Use the worked results to check what each formula evaluates.

CLOSE

The report starts with comparable results

April YoY

+58.2%

Jan–Aug YoY

+27.1%

Order 100142

94.00

Check the dates, the rows and the filters. Next, add interactions and tell the story behind the numbers.